

COURSE DESCRIPTION

Course Number	0418-492	Course Title	Capstone Project II
Credit Hours	3 (3:0:3)	Course Coordinator	Capstone Project Committee
Compulsory	Yes	Prerequisite	0418-491

Description

This course is taken by senior students under the guidance of a faculty member. By the time of enrolling in this course, students are expected to have their project proposals presented and approved. Capstone Project II aims to provide students with a capstone design and development experience. It enables students to demonstrate and enhance their practical skills such as developing system requirements specifications, learning and using industry standard computing environments, exploring new demanding application domains, and building executable prototypes. Students will also experience significant scientific and technical writing, working in teams, interacting with users, and making oral presentations. Enrolled students must develop software preferably using a new environment, submit a comprehensive project report and a fully functional system, and publicly present their work.

Textbook

Title	Essential Scrum: A Practical Guide to the Most Popular Agile Process
Author	Kenneth S. Rubin
Publication	First Edition, Addison-Wesley, 2012
ISBN-10	0137043295
ISBN-13	9780137043293

Major Topics Covered in the Course

No.	Topics	Hours
1	Professional and technical writing	3
2	Agile Principles	3
3	Scrum Framework	6
4	Requirements and User Stories	3
5	Product Backlog, Estimation and Velocity, technical debt	6
6	Scrum Planning and Risk Management	3
7	Presentations	6
8	Selected Topics	9
9	Exams	3
	Total	42

Grading

Item	Weight	Provided by
Team Performance	10	Course Instructor
Intermediate Reports	20	Course Instructor
Final Presentation	10	Capstone Committee
Final Report	20	Capstone Committee
Final Executable Software	40	Capstone Committee

Course Outcomes

Students should be able to:

1. Demonstrate effective and professional collaboration in a team.
2. Produce an intermediate and a final report describing the analysis, modeling, and testing of a complete software product.
3. Produce a complete executable software product based on requirements approved by the capstone project committee.
4. Understand and explain agile software project management principles and practices.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1			M	H	H	
2	M		H		H	M
3	H	H			H	H
4	H					

Last updated on Jan 8, 2025.

Approved by the capstone and curricula committee on Jan 8, 2025.